

NCEMS – CURE: Project Outline

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(I) Examining Two “Shot-Gun” Proteomics Experiments

Shotgun proteomics is a high-throughput strategy for identifying and quantifying proteins in complex biological samples. It works by enzymatically digesting proteins into peptides, separating those peptides via liquid chromatography, and analyzing them using tandem mass spectrometry (MS/MS). This approach allows researchers to characterize thousands of proteins in a single experiment without prior knowledge of the sample’s composition.

In this project, we will explore how protein abundance information can be extracted from public datasets in the [PRIDE Archive](#), using mass spectrometry-based shotgun proteomics. We focus on two biologically and technically distinct datasets:

- Soybean (*Glycine max*): [PXD023343](#)
- Human (*Homo sapiens*): [PXD005187](#)

These examples illustrate different use cases, labeling strategies, and experimental goals in proteomics.

(I – 1) Extracting key information from the publications

In this section, you’ll extract important metadata from both the original publication and the experimental files associated with your dataset. This information will be used to build a *.json* configuration file required by SAGE to process the data.

[1] DOI: 10.1038/cddiscovery.2017.10, Des: Kathiriyaa JJ, Nakra N, Nixon J, Patel PS, Vaghasiya V, Alhassani A, Tian Z, Allen-Gipson D, Davé V. Galectin-1 inhibition attenuates profibrotic signaling in hypoxia-induced pulmonary fibrosis. Cell Death Discov. 2017 Apr 10;3:17010. eCollection 2017

[2] DOI: 10.3390/ijms22020920, Des: Cheng L, Min W, Li M, Zhou L, Hsu CC, Yang X, Jiang X, Ruan Z, Zhong Y, Wang ZY, Wang W. Quantitative Proteomics Reveals that GmENO2 Proteins Are Involved in Response to Phosphate Starvation in the Leaves of Glycine max L. Int J Mol Sci. 2021 22(2)

Q. What cleavage enzyme was used and what amino acids does it cut after?

Why it matters: The enzyme determines the peptide cleavage rules which are crucial for understanding how a protein will fragment in the spectrometer.

Where to look: Check the methods section of the manuscript for sample preparation or digestion steps. Common entries include trypsin, Lys-C, chymotrypsin, or Glu-C. This information may also be in PRIDE metadata under “Sample Protocol” or “Digestion.”

Table 1: Common Cleavage Enzymes

Enzyme	Cleaves After (Residues)	Cleavage Blocked By	Terminal Side of Cleavage	Notes
Trypsin	K, R	P	C-terminal	Most commonly used; highly specific
Lys-C	K	—	C-terminal	Stable in high denaturant (e.g., urea)
Arg-C	R	—	C-terminal	Less specific than trypsin
Glu-C	E, sometimes D	—	C-terminal	Cleaves more broadly at high pH
Chymotrypsin	F, Y, W, L, M	P	C-terminal	Broader specificity; more missed cleavages likely
Asp-N	D	—	N-terminal	N-terminal cleavage; less common
Pepsin	Broad (especially F, L, E)	pH-dependent	C-terminal or mixed	Non-specific, active at low pH
Thermolysin	L, I, V, F, M	—	N-terminal	Metalloprotease, stable at high temperature

Q. Was the dataset labeled or label-free?

Why it matters: Labeling strategies (e.g., SILAC, TMT) enable multiplexed, highly accurate quantification, while label-free methods are simpler and more scalable. Knowing the approach clarifies the strengths, limitations, and comparability of the quantitative results.

Where to look: Look for keywords like “TMT”, “iTRAQ”, or “LFQ” in the methods section or PRIDE metadata.

TMT-Labeled Data

TMT (Tandem Mass Tags) are chemical labels attached to peptides that allow multiplexed MS-based quantification using reporter ions. These are detected at MS2 or MS3 levels.

Use this when:

- The methods mention TMT10, TMT11, TMT16, or iTRAQ.
- Quantification is based on reporter ion intensities.

SILAC-Labeled Data

SILAC (Stable Isotope Labeling by Amino acids in Cell culture) incorporates heavy isotopes into amino acids (e.g., $^{13}\text{C}_6^{15}\text{N}_4\text{-Arg}$ & $^{13}\text{C}_6^{15}\text{N}_2\text{-Lys}$) during protein synthesis, shifting precursor m/z but not fragment ions.

Use this when:

- The methods mention *SILAC*, *light/heavy*, or *metabolic labeling*.

Label-Free Quantification (LFQ)

LFQ uses *precursor ion intensity* without any chemical or isotopic labeling. It's based on matching peptide features across runs.

Use this when:

- There is *NO mention of TMT, iTRAQ, or SILAC, sometimes the authors will explicitly state they used LFQ methods.*

Q. Were any specific chemical modifications or post-translational modifications (PTMs) mentioned by the authors?

Why it matters: These modifications often change the mass and charge of a peptide and thus its mass spectra. Sometimes these modifications are intentional such as in the case of labeling experiments to test how protein abundances changes under different conditions.

Where to look: Usually listed under *sample preparation* or *search parameters* in the methods section.

Static vs. Variable Modifications

In proteomics, modifications refer to changes in the mass of a residue or terminus caused by chemical labeling, post-translational modifications (PTMs), or experimental artifacts.

Static Modifications

A *static modification* is a change that occurs on every instance of a given residue or terminus.

Use static mods when:

- The modification was applied to all peptides during sample preparation
- You are using chemical labels like TMT or iTRAQ
- The mass shift is consistent and universal

Examples of common static mods:

Modification	Residue	Mass Shift	When It Occurs
Carbamidomethylation	C	+57.0215	During alkylation with iodoacetamide
TMT16 labeling	K, ^	+304.207	Chemical labeling (TMT 16plex)
iTRAQ 8plex	K, ^	+304.2054	iTRAQ chemical labeling

Variable Modifications

A *variable modification* is optional — the search engine will consider both the modified and unmodified versions of the peptide. This allows detection of PTMs or partial labeling.

Use variable mods when:

- You expect the modification to occur on only some peptide
- You're searching for biologically relevant PTMs
- You want to allow missed modifications (e.g., incomplete labeling)

Examples of common variable mods:

Modification	Residue	Mass Shift	When It Occurs
Oxidation	M	+15.9949	Spontaneous or regulated (PTM)
Deamidation	N, Q	+0.984	Non-enzymatic or enzymatic (PTM)
Pyro-glutamate formation	^Q, ^E	-17.0265 / -18.0106	N-terminal loss of ammonia
Acetylation	^	+42.0106	N-terminal (PTM)
SILAC heavy lysine (labeling)	K	+8.0142	Metabolic labeling
SILAC heavy arginine	R	+10.0083	Metabolic labeling

(II): Open Modification Search with Spectral Alignment Guided Engine (SAGE)

Why this pipeline?

We need one workflow that (a) finds what peptides are present in the mass spec files even if they carry unexpected chemical modifications, and (b) turns those signals into numbers we can compare across runs/conditions. FragPipe is a highly used and gold standard pipeline that delivers both.

What the pipeline does (at a glance)

- I. Scans with a wide window to catch peptides that may have unknown/variable modifications (shows up as mass shifts).
- II. Controls error rates and maps peptides to their most likely parent protein.
- III. Estimates the peptide abundances

(II – 1) Examine the peptide quantification file

Let us examine the peptide level protein mapping and abundance estimation from Fragpipe for either the Soybean or Human samples. You can find them located in:

- data/Soybean_msstats.csv
- data/Human_msstats.csv

Each of these files contains one peptide fragment per row that was observed and confidently matched and quantified in the mass spectrometry experiments pipeline.

Fragpipe Output File Description

Column name	Description
ProteinName	The protein accession(s) the peptide maps to, in UniProt-style format (e.g., sp P12345 PROT_NAME).
PeptideSequence	The amino-acid sequence of the identified peptide (after in silico digestion, e.g., tryptic).
Protein.Start	Start residue position of the fragment in the Uniprot canonical sequence
Protein.End	End residue position of the fragment in the Uniprot canonical sequence
PrecursorCharge	The precursor ion charge state observed in MS1 (e.g., +2, +3).
IsotopeLabelType	L = light sample with no isotope labeling H = heavy sample with isotope labeling
condition	Experimental condition or sample group (e.g., treatment, control).
BioReplicate	Biological replicate identifier for the sample.
Run	The specific LC-MS/MS run or file name.
Intensity	Quantified signal intensity for the peptide or fragment ion in the run.

Q. Why do some peptides have only letters and others have numbers in brackets?

- TAEENLDR vs. EM[+15.9949]NDAAMFYTNR for example

Q. What is the range, mean, median, and mode of the peptide charges observed?

Q. (Bonus) why do we ignore +1 charged peptide fragments?

Q. Do any peptides map to more than 1 protein?

Q. What is the difference between a q-value reported here and a p-value?

Q. What are the units of the spectral intensity (if any)?

(III) Calculate and Examine the Protein Abundance Distributions

Now that you have examined the list of identified peptides that have been confidently matched to proteins of interest we need to obtain estimates of the protein level abundances. Here we will lay out a

general procedure for this quantification so you can attempt it on your own using your preferred programming language. You are also free to use the python Jupyter notebooks/scripts located in the `src/data/` folder.

(III – 1) Filter the proteins by coverage

Explanation: Filters out low-coverage proteins where less than 50% of their structure was observed.

Why: Not all proteins are present in the cell at the same concentration. As with most spectroscopic measurements the signal observed in mass spectrometry proteomics experiments is roughly proportional to the population of the protein in the sample. Every instrument has a limit of detection where if the signal is too weak/strong it will no longer be recorded. Many proteins may fall near or below this threshold and thus have much greater uncertainty in their acquisition. As a simple filter we remove proteins whose sequences were not fully resolved.

Note: The threshold chosen depends on how complex your sample is. If you are working with purified protein sample only containing a single protein, then you can choose a threshold of up to 95% for the coverage. But as your sample gets more complex and heterogeneous then the you need to relax the threshold as your signal gets more and more segmented and only parts of proteins are ever resolved with certainty.

(III – 2) Estimate to protein-level abundance per run and compute $\log_2$ fold-change

Explanation:

- Estimate protein abundance by summing over the peptide abundances for that protein in each technical replicate and condition.
- Handle zeros via an Limit of Detection (LOD) based gaussian imputation: for any 0 in the dataset substitute a small random value drawn from a narrow Gaussian centered on a fixed LOD. This preserves rank order while avoiding division-by-zero and infinite ratios.
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Why: Summing to protein-level per run aligns with standard bottom-up quantification, reducing peptide-level noise and simplifying downstream stats. Pivoting creates paired condition columns for stable ratio calculations. LOD imputation prevents undefined ratios and overly-inflated fold-changes caused by zeros, yielding interpretable $\log_2$ contrasts suitable for averaging across runs or statistical testing.

Note: Alternatives include mean, weighted mean (by peptide quality), or more sophisticated models (like MaxLFQ's ratio-based inference). Students can compare results of median vs mean as an exploration. Some further reading on how complicated this abundance estimation can get:

<https://www.nature.com/articles/nmeth.3901.pdf>

(III – 3) Compute Sample Statistics and Make Box and Violin Plots of each Sample Separately

You should compute the following statistics for each sample:

1. The estimated protein abundance mean
2. The estimated protein abundance minimum and maximum
3. The estimated protein abundance median
4. The estimated protein abundance first and third quartiles
5. The estimated protein abundance variance

Q. Do you observe any noticeable differences in the distribution of protein abundances between technical replicates under the same condition?

Q. How about between conditions?

Q. What does this tell you about whether the treatment had an effect?

Q. Is it possible that there is affect we cannot observe by examining this population level change? If so how?

(IV): Differential Protein Abundance from LFQ Mass Spectrometry Data Datasets

(IV – 1) Statistical testing (Welch's t-test)

Explanation: Apply Welch's t-test per protein, comparing the standardized abundance vectors between the two conditions.

Why Welch's test? It does not assume equal variance between groups, a prudent choice for biological data where heteroskedasticity is common.

Note: There are only 3 replicates per group and degrees of freedom are low which typically leads to a high variance.

Note: Welch's t-test is generally powerful when sample sizes are modest and distributions are approximately symmetric; the Mann–Whitney test is robust to non-normality but tests for difference in distribution location in a different way. It's good practice to examine data (e.g., via violin/box plots) to decide if assumptions are reasonable.

(IV – 3) Ranking and multiple testing correction

Explanation: Sorts proteins by raw significance (smallest p-value first) to prioritize candidates.

- Apply Benjamini–Hochberg false discovery rate (FDR) correction to control the expected proportion of false discoveries among proteins.

Why: Thousands of proteins may be tested; without correction, the number of false positives at $p < 0.05$ would be unacceptably high. FDR balances discovery with error control.

Note: There are different FDR correction methods that rely on different assumptions. You should always check that the assumptions are not violated in your dataset before application of any statistical method.

Q. What fraction of proteins had a significant change in their abundance?

(IV – 4) Volcano plots

A volcano plot is a scatter plot that simultaneously displays both magnitude of change and statistical significance for each protein (or peptide) when comparing two conditions. It's a rapid visual filter to highlight candidates that are both substantially and confidently different.

Axes of a volcano plot:

- X-axis: Fold change: This is usually reported as the, $\log_2(\text{treatment/control})$, of the abundance ratio between the two conditions (positive means higher in treatment condition, negative means higher in control condition). It reflects effect size how much a protein's abundance changes.
- Y-axis: $-\log_{10}(\text{adjusted p-value})$ from a statistical test (e.g., Welch's t-test on replicate abundances). Higher values mean stronger statistical evidence against the null (i.e., more significant). Using the negative log makes small p-values (strong significance) appear at the top.

Q. How many proteins have a significant change after treatment and a $\log_2(T/C) \leq -1$?

Q. How many proteins have a significant change after treatment and a $\log_2(T/C) \geq 1$?

Q. How many proteins have a significant change after treatment and a $-1 \leq \log_2(T/C) \leq 1$?

Q. How many proteins DO NOT have a significant change after treatment and a $\log_2(T/C) \leq -1$?

Q. How many proteins DO NOT have a significant change after treatment and a $\log_2(T/C) \geq 1$?

Q. How many proteins DO NOT have a significant change after treatment and a $-1 \leq \log_2(T/C) \leq 1$?

Q. which of these groups would you focus your efforts on investigating further to understand how the treatment is potentially affecting protein production and function?